

Kakarla Venkata Rama Sai Pavan Shanmukh

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Summary

Adaptable B.Tech student and full-stack developer specializing in scalable distributed ledger analytics and secure real-time systems to drive data-driven architectural innovation.

Education

KLEF Deemed to be University, Vijayawada | 9.56 CGPA Aug 2023 - Present

Bachelor of Technology (BTech) | Computer Science and Information Technology

Narayana Educational Institutions, Vijayawada | 93.1% Aug 2021 - Jul 2023

Intermediate | Andhra Pradesh Board of Intermediate Education

Nalanda Vidya Niketan, Vijayawada | 81.8% Jul 2020 - May 2021

10th Grade | Central Board of Secondary Education

Experience

Cisco Networking Academy May 2024 - Jul 2024

Virtual Internship

Hands-on networking with Cisco Packet Tracer. Designed and simulated network topologies; implemented and tested routing algorithms for optimized performance.

AICTE May 2025 - Jul 2025

Web Developer Virtual Intern

Developed and deployed full stack web applications using ReactJS, Spring Boot, and Node.js. Integrated MySQL database with secure authentication and OTP-based login. Implemented RESTful APIs and followed clean architecture principles

Skills

Core: Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks, Linux, OOPS Through Java, Git and GitHub

Languages: Java, Python, HTML5, Linux, CSS3, JavaScript

Tools, Frameworks and Databases: React.js, Spring Boot, Node.js, Power BI, CI/CD pipelines, MySQL, PostgreSQL

Projects

[SentinelStream: Real-Time Ransomware Detection via Behavioral Anomaly Monitoring](#) Feb 2026 - May 2026

Abstract: A real-time Watchdog-based behavioral monitoring and threshold logic to identify ransomware encryption with 80% accuracy.

Technologies, Tools & Software: Python, Watchdog, Flask, MySQL, REST APIs, and behavioral anomaly monitoring.

Outcome and Social Relevance: Minimized detection latency to 10 seconds & safeguarding critical infrastructure against data loss.

Technologies: Python, React.js, Network Security, JavaScript, Git, GitHub

[OmniBank: Full-Stack Financial Management System with Automated CI/CD](#) Oct 2025 - Jan 2026

Abstract: A secure full-stack banking platform featuring real-time transactions, MFA, and Automated CI/CD pipelines.

Technologies, Tools & Software: React.js, Vite, Spring Boot, Spring Security, MySQL, REST APIs, and CI/CD tools.

Outcome and Social Relevance: Provided a platform for transparent financial tracking.

Technologies: JavaScript, CI / CD, Docker, Kubernetes, Jenkins

[LexiTree: Optimized Text Analysis via Self-Balancing AVL Trees](#) Nov 2023 - Dec 2023

Abstract: Decreased AVL Tree's performance to O(log n) word frequency analysis & visualization for large-scale textual datasets.

Technologies, Tools & Software: AVL Tree, UI Frameworks, and Performance Analysis modules.

Outcome and Social Relevance: Optimized large-scale data processing.

Technologies: React.js, Web Technologies, Data Structures and Algorithms, Git, GitHub

Certifications and Achievements

Microsoft Certified Azure Fundamentals	Mar 2026
Acquired basic knowledge of cloud services and how they are provided via Microsoft Azure.	
Optimized Antenna Design for WBAN Applications	Dec 2024 - Present
A high-efficiency, flexible design featuring a metamaterial-backed ground plane to ensure stable on-body communication.	

Coding Profiles

Leetcode: [klu2300090201](#) Codechef: [kakarla_pavan1](#) Codeforces: [2300090201](#)